

UNIVERSITY OF ASIA PACIFIC
DEPARTMENT OF CSE

Lab Report - 6

COURSE CODE: CSE 402.

COURSE TITLE: OPERATING SYSTEM LAB.

SUBMITTED BY:

NAME : Morium Hossain Mim

STUDENT ID : 23101134

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SUBMITTED TO

ATIA RAHMAN ORTHI

LECTURER AT UAP

1. Problem:

Given the following **disk request queue** and the **initial head position**:

Parameter	Value
Disk Request Queue	0, 30, 41, 62, 14, 20
Initial Head Position	20

Implement the following Disk Scheduling Algorithms:

- **FCFS (First Come First Served)**
- **SSTF (Shortest Seek Time First)**

For each algorithm, calculate the **Total Seek Time** (total head movement).

Finally, display the results and **compare the Total Seek Time of FCFS and SSTF** to determine which algorithm performs better for the given requests.

2. Source Code:

FCFS & SSTF disk scheduling

```
req  = [0, 30, 41, 62, 14, 20]
head = 20


req = [r for r in req if r != head]

# FCFS
total_fcfs = 0
cur = head
for r in req:
    total_fcfs += abs(r - cur)
    cur = r

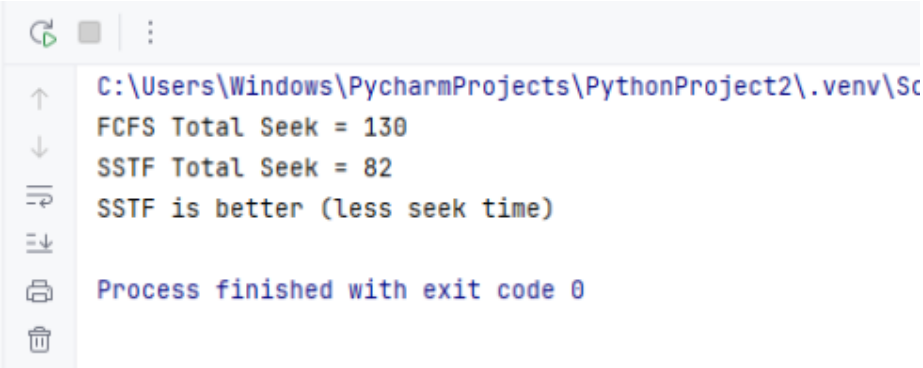
# SSTF
remaining = req.copy()
total_sstf = 0
cur = head
while remaining:
    closest = min(remaining, key=lambda x: abs(x - cur))
    total_sstf += abs(closest - cur)
    cur = closest
    remaining.remove(closest)

print("FCFS Total Seek =", total_fcfs)
print("SSTF Total Seek =", total_sstf)

if total_sstf < total_fcfs:
    print("SSTF is better (less seek time)")
else:
    print("FCFS is better (less seek time)")
```

3. Output:

FCFS & SSTF disk scheduling



4. Discussion:

In this experiment, **FCFS** and **SSTF** disk scheduling algorithms were implemented using the same request queue and the same initial head position (**20**). Their performance was compared using the **Total Seek Time** (total head movement).

The request for cylinder 20 is the same as the initial head position, so it needs no head movement and was removed from the queue. The remaining requests are **0, 30, 41, 62, 14**.

For **FCFS**, the requests are served in the order they arrive: 20 → 0 → 30 → 41 → 62 → 14. The head movements are 20 + 30 + 11 + 21 + 48, so the Total Seek Time was **130**. The head moves back and forth a lot, which increases the seek time.

For **SSTF**, the request closest to the current head position is served first: 20 → 14 → 0 → 30 → 41 → 62. The head movements are 6 + 14 + 30 + 11 + 21, so the Total Seek Time was **82**.

From the results, **SSTF** reduced the head movement by 48 cylinders (about 36.9%) compared to FCFS. Hence, for the given set of requests, SSTF provided better performance.

5. Conclusion:

The experiment demonstrated the implementation and performance comparison of FCFS and SSTF disk scheduling. The results show that **SSTF** performed more efficiently for the selected requests, with a Total Seek Time of **82**, while FCFS had a Total Seek Time of **130**.

However, SSTF may cause starvation for requests that are far from the current head position, whereas FCFS is simple and fair because it serves requests in arrival order.

6. GitHub:

FCFS & SSTF disk scheduling

https://github.com/23101134Mim/CSE_402_Operating_System_Lab/tree/main/FCFS%20%26%20SSTF%20disk%20scheduling